Rank-27 Vivify Follow-up: Exact Sweep/Substitute Retirement Order

Tactix engineering handoff

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Executive outcome

The earlier Vivify variable-range cutoff was indeed unsupported and remains removed. The next rank-27 divergence was not another Vivify heuristic. The two solvers had identical EVSIDS scores and retired the same 369 variables, but Tactix removed those variables from its equal-score heap in environment batch order. Kissat's source delays deactivation until the single Substitute call after Sweep and scans the complete union in compact internal-variable order. That order difference changed heap topology and selected a different 35th decision after Probe.

Tactix now preserves the same transaction. Ordinary Substitute emits retirements in permanent first-import order; Sweep normalizes only the suffix it produced; DefaultPolicy extends that suffix through the mandated post-Sweep Substitute so any additional SCC variables join the same scan. Vivify's earlier retirement prefix stays ahead of it. The repair contains no formula range, instance identity, rank, hash, literal, deadline, or tuned constant.

The repaired candidate matches all 369 retained Kissat heap removals exactly, including scores and heap positions, and all 481 decisions through the next conflict. Release passes 72/72. The full 28B+5C guard passes 66/66 with every model and proof checked and zero residue. Immutable rank 27 remains UNKNOWN at 1,200.004 seconds; its partial proof was hashed and deleted, so rank 27 remains open without certificate residue.

Why this appeared to be Vivify

The common state sits after two Vivify calls, TransitiveReduce, Sweep, and its follow-up Substitute. Vivify's terminal effects had already been audited: all 69 binary-producing probe effects matched. Nevertheless, the first search decision difference appeared immediately after this combined Probe pipeline, so a candidate-range or clause-order explanation initially looked plausible.

A filtered heap trace separated formula effects from representation. After 34 shared decisions, Kissat's heap root was external variable 230665 and Tactix's was 230802, but both scores were exactly `2.6056478753115483e+24`. Thus score bumping was not the cause. Both traces then showed 369 retired variables with identical sets. The first 26 heap removals were byte-for-byte equal; removal 27 was variable 95250 in Kissat and 95251 in Tactix. Equal-score binary heaps preserve this mutation history, so different removal order is enough to change the root.

The source-backed lifecycle

Kissat's `sweep.c` proves an equivalence, rewrites the retiring literal's connected clauses, and records representatives in `sweeper->reprs`. It does not deactivate the variable. The fixed Probe pipeline then calls `kissat_substitute`; its representative-removal pass walks `for (all_variables(idx))` and deactivates the entire SCC union. Kissat's compact internal indices correspond to Tactix's permanent first-import order.

Tactix must create a reconstruction witness as each inline environment rewrites its variable-width arena. That representation requirement does not authorize the environment boundaries to become heuristic scheduling state. The first repair attempt made this distinction clear: ordering each batch by equivalence emission still began `95251, 420557, ...`, while Kissat began `95250, 95251, 95252, ...`. Ordering within each environment by import rank then matched removals 1–363, but four Sweep variables still preceded two SCC variables discovered only by the follow-up Substitute. Carrying the suffix boundary across that follow-up reproduces Kissat's one union.

Exact diagnostic evidence

Candidate-only job `1637838` used the retained Kissat trace; no reference solver was rerun. Each heap-removal payload records the removed external variable and exact score, its old position, the swapped tail variable, the heap size before removal, and the tail's final position.

Boundary	Tactix	Kissat artifact	Comparison
Retirement set	369	369	identical
Retirement payloads	369	369	exact order/content
Payload digest	`ad1a5189...34f1a7`		identical
Conflict-3069 decisions	481	481	exact order
Decision digest	`fd9a9884...2b6f98`		identical

The former 35th-decision split is closed: both now choose external literal `-230665`. Focused tests separately pin a numeric-versus-import-order SCC and two productive Sweep environments forced in reverse import order. The latter assertion fails before the transaction-boundary normalization.

Full validation guard

Matched campaign `benchmarks/runs/20260821-1215-BC-sweep-substitute-retirement/`, job `1637839`, ran all 28 Tier-B fixtures and the frozen five mechanism-relevant Tier-C fixtures under 16 GiB and 1,200 seconds.

Arm	valid	SAT / UNSAT	aggregate	peak RSS
Accepted pin	33/33	17 / 16	2212.751 s	3.233 GiB
Candidate	33/33	17 / 16	3071.347 s	2.969 GiB

All 34 SAT models and 32 UNSAT proofs verify. There are no solve losses, invalid or malformed results, checker failures, cleanup failures, or residual certificates. The candidate ratio is 1.388022, so this is a correctness iteration guard rather than a new accepted performance pin. Candidate `T105.2.0` proves UNSAT in 863.983 seconds versus 794.595 seconds for the accepted pin; both proofs were checked and deleted.

Immutable witness and next action

Only after the complete guard passed was immutable witness job `1637906` submitted under 16 GiB and 1,200 seconds. It returned UNKNOWN at 1,200.004 seconds. The runner recorded a 132,120,576-byte partial proof, SHA-256 `7f193dcb...ab73fc6`, then deleted it and all derived scratch. Certificate residue is zero. Rank 27 is not added to the nine closed ranks.

The next rank-27 localization, if needed, starts after the now-exact conflict-3069 heap mutations and 481-decision prefix. It must retain the closed Vivify scope/deduction/interleaving, TransitiveReduce order, Sweep semantic dose and representation lifecycle, exact seed/equivalence schedule, and this Sweep-to-Substitute retirement transaction. Reintroducing a Vivify variable-range policy would contradict both source and the localized evidence.

The durable technical report is `benchmarks/reports/2026-08-21-tierd-rank27-sweep-substitute-retirement.md`. The resume surface is `PROGRESS.md`.